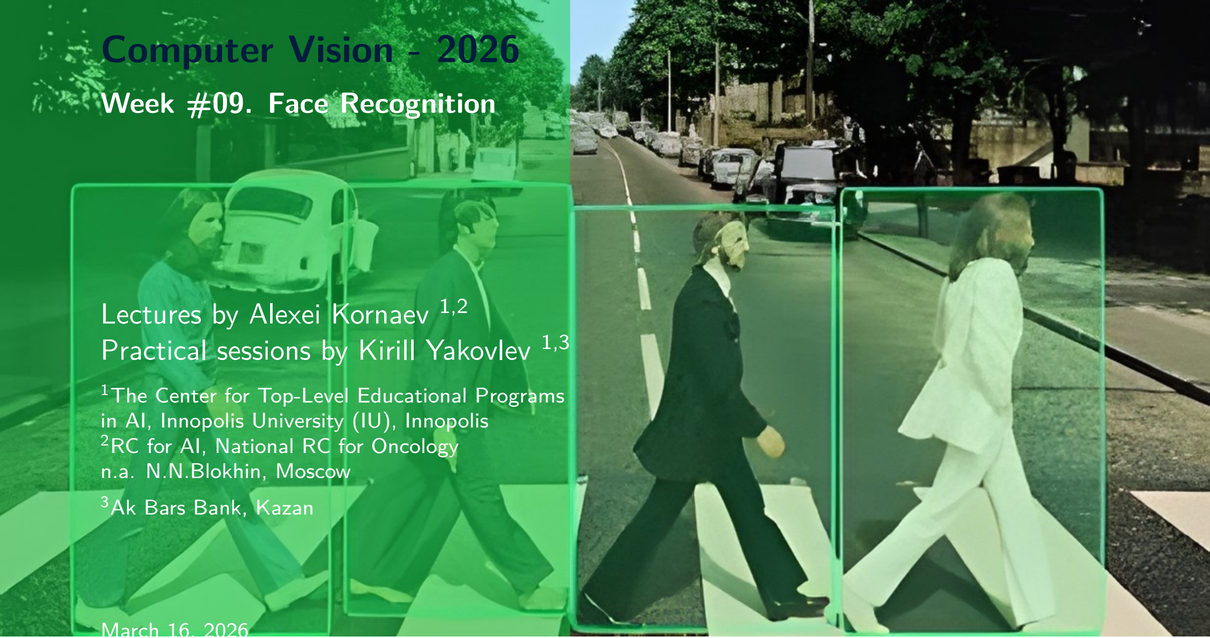

Computer Vision - 2026

Week #09. Face Recognition

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Agenda

Section 1. Outcomes

Outcomes

By the end of this week, students will be able to:

1. Distinguish between **face verification** (1:1 matching) and **face identification** (1:N search).
2. Explain the modern face recognition pipeline: **detection/alignment** $\rightarrow$ **embedding model** $\rightarrow$ **similarity scoring** $\rightarrow$ **thresholding/search**.
3. Compare major loss families for learning face embeddings: **contrastive loss, triplet loss, and angular-margin losses** such as ArcFace.
4. Understand why **image quality** matters and how modern methods such as quality-aware margins improve robustness.
5. Evaluate a face recognition system using **ROC**, **TAR/FAR**, **EER**, and **Rank-1 / CMC**, and discuss practical trade-offs.
6. Discuss limitations and risks related to pose, occlusion, domain shift, fairness, privacy, spoofing, and deepfakes.

Key takeaway: Modern face recognition is primarily an **embedding learning + similarity decision** problem, where data quality, evaluation protocol, and deployment constraints matter as much as raw accuracy.

Modern Face Recognition Pipeline

Core idea

Modern face recognition systems do not usually predict an identity directly. Instead, they learn a compact embedding where images of the same person are close and different identities are far apart.

1. **Face detection and alignment:** detect the face and normalize geometry.
2. **Embedding model:** encode the aligned face into a feature vector.
3. **Normalization:** often use L_2-normalized embeddings.
4. **Similarity scoring:** cosine similarity or Euclidean distance between embeddings.
5. **Decision:**
   - **Verification:** compare score with a threshold.
   - **Identification:** search in a gallery and return top-k matches.

Training vs inference:

- During training: margin-based metric/classification losses shape the embedding space.
- During inference: similarity scoring and threshold selection determine behavior.

Section 2. Challenges in FR

Challenges in Face Recognition

Overview of Challenges

Face recognition systems face several challenges that can significantly impact their performance. These include pose variation, illumination changes, occlusion, ethical concerns, and adversarial attacks.

- **Pose Variation:** Faces can appear in different orientations, making it difficult to match them.
- **Illumination Changes:** Variations in lighting conditions can alter the appearance of faces.
- **Occlusion:** Parts of the face may be obscured by objects or other faces.
- **Ethics:** Privacy concerns and biases in face recognition systems.
- **Adversarial Attacks:** Deliberate attempts to fool face recognition systems.

Pose Variation

Pose Variation

Pose variation refers to the different orientations of a face relative to the camera. This can include frontal, profile, and oblique views.

- **Impact:** Reduces the accuracy of face recognition systems.
- **Solutions:** Use of 3D models, multi-view training data, and pose-invariant features.

Illumination Changes

Illumination Changes

Illumination changes refer to variations in lighting conditions that can alter the appearance of a face.

- **Impact:** Can cause significant changes in the appearance of facial features.
- **Solutions:** Normalization techniques, use of infrared imaging, and deep learning models trained under various lighting conditions.

Occlusion

Occlusion occurs when parts of a face are obscured by objects (e.g., masks, sunglasses), other people, or environmental factors. It degrades recognition accuracy by hiding critical facial features.

Understanding Occlusion

Figure: Examples of occluded face images from the MAFA dataset [Zeng et al., 2021].

- **Impact:**
  - Loss of key features (e.g., eyes, mouth).
  - State-of-the-art models like DeepFace or ArcFace suffer >30% accuracy drop under heavy occlusion.
- **Common Occlusions:** *Masks* (e.g., medical masks). *Accessories* (e.g., sunglasses, scarves). *Environmental* (e.g., hands, hair).
- **Challenges:**
  - Partial face data limits recognition methods.
  - Dynamic occlusions require real-time adaptation.

Solutions for Occlusion Handling

Technical Approaches

Addressing occlusion requires robust feature extraction and reconstruction techniques.

- **Partial Face Recognition:**
  - Train models on cropped facial regions (e.g., eyes-only, mouth-only). Use *metric learning* to match partial and holistic embeddings.
- **Attention Mechanisms:**
  - Focus on visible regions, e.g. *Transformer Networks*) may prioritize non-occluded areas with attention [Liu et al., 2023].
- **Generative Models:**
  - Reconstruct occluded regions using GANs, Diffusion models.

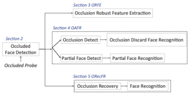

Figure: FR under occlusion [Zeng et al., 2021].

Responsible Deployment: Fairness, Privacy, Regulation

Important distinction

A benchmark score does not automatically justify deployment in a real system.

- **Fairness:** error rates may differ across demographic groups, capture conditions, and thresholds.
- **Privacy:** face data are biometric data and require strict handling, storage, and consent policies.
- **Evaluation discipline:** claims about fairness or robustness should be based on protocol-specific audits, not only on one aggregate accuracy number.
- **Regulation:** biometric applications are increasingly governed by explicit legal constraints and risk categories.

Teaching point:

- what the model predicts,
- how the system is evaluated, and
- whether the deployment is ethically and legally acceptable.

Security Threats in Face Recognition

Beyond ordinary test error

A face recognition system may fail not only because of natural variation, but also because of deliberate attacks.

- **Presentation attacks / spoofing:** printed photos, replayed videos, masks.
- **Morphing attacks:** manipulated images designed to match multiple identities.
- **Adversarial perturbations:** inputs modified to change matching behavior.
- **Model inversion / privacy leakage:** attempts to recover sensitive information from model outputs or parameters.

Defenses

- liveness / presentation-attack detection,
- robust training and data hygiene,
- threshold selection under realistic attack models,
- privacy-aware system design.

Adversarial Attacks: Threat Models

Adversarial Attacks in Face Recognition

Adversarial attacks manipulate inputs to deceive face recognition systems. Common types include:

- **Evasion Attacks:**
  - *Goal:* Fool the system during inference (e.g., adversarial glasses).
- **Poisoning Attacks:**
  - *Goal:* Corrupt training data (e.g. injecting malicious samples).
- **Model Inversion Attacks:**
  - *Goal:* Reconstruct training data from model outputs (privacy breach).

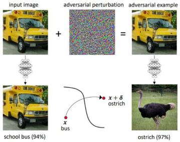

Figure: Adversarial noise attack.

Defending Against Adversarial Attacks

Robustness Strategies

Mitigating adversarial attacks requires proactive defenses and robust model design.

- **Adversarial Training:**
  - Train models on adversarial examples to improve robustness.
  - *Limitation:* Computationally expensive; does not generalize to all attacks.
- **Input Preprocessing:**
  - Apply noise reduction, quantization, or JPEG compression to inputs.
- **Detection Mechanisms:**
  - Deploy detectors to flag adversarial inputs (e.g., using inconsistency in feature space).
- **Certified Defenses:**
  - Use mathematically proven robust models (e.g., randomized smoothing).

Section 3. Traditional FR Methods

Traditional Face Recognition Methods

Early approaches relied on dimensionality reduction and handcrafted features.

Linear Methods

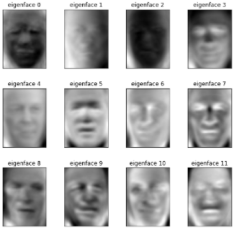

Figure: Eigenfaces [Turk and Pentland, 1991].

- **Eigenfaces [Turk and Pentland, 1991]:**
  - Uses PCA to project faces into a low-dimensional subspace.
  - Maximizes variance to capture dominant facial features.
- **LBPH (Local Binary Patterns Histogram):**
  - Encodes texture patterns using local binary operators.
  - Robust to monotonic illumination changes but struggles with pose/occlusion.

Section 4. Deep Learning for FR

Deep Embedding Models for Face Recognition

Modern face recognition models learn discriminative face embeddings and compare them with similarity scores.

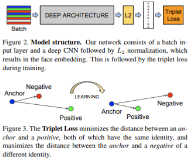

Figure: Face embedding intuition [Schroff et al., 2015; Deng et al., 2019].

Main families

- **Siamese / contrastive models [Koch et al., 2015]:** learn similarity from positive and negative pairs.
- **Triplet-based models (FaceNet) [Schroff et al., 2015]:** enforce anchor-pos. pairs to be closer than anchor-neg. pairs by a margin.
- **Angular-margin softmax models (ArcFace) [Deng et al., 2019]:** optimize class separation directly on a normalized hypersphere.

Current practice:

- normalized embeddings;
- cosine similarity at inference;
- margin-based classification losses for scalable training.

Why ArcFace-style losses became dominant: they are usually easier to optimize at large scale than explicit pair/triplet mining while still producing strong embeddings.

Quality-Aware Face Embeddings

Why quality matters

Low-quality faces (blur, low resolution, bad lighting, extreme pose, occlusion) are much harder to recognize reliably than clean frontal images.

- **MagFace [Meng et al., 2021]:** learns embeddings whose magnitude is related to face quality, helping the model organize easy and hard samples more meaningfully.
- **AdaFace [Kim et al., 2022]:** adapts the margin according to image quality, improving recognition on degraded and unconstrained imagery.
- **Practical lesson:** modern FR is not only about identity separation, but also about estimating when the input is reliable enough for matching.

Loss Functions in Face Recognition

Metric Learning Objective

A loss enforces that embeddings of the same identity are closer than those of different identities.

- **Contrastive Loss :**

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{N} y_i \cdot d_i^2 + (1 - y_i) \cdot \max(\alpha - d_i, 0)^2$$

where $y_i = 1$ for same-class pairs, d_i=embedding distance, α=margin.

- **Triplet Loss [Schroff et al., 2015]:**

$$\mathcal{L} = \sum_{i=1}^{N} \max \left(||f(a_i) - f(p_i)||^2 - ||f(a_i) - f(n_i)||^2 + \alpha, 0 \right)$$

- **ArcFace Loss [Deng et al., 2019]:**

$$L = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos(\theta_j)}}$$

where s=scaling factor, m=angular margin, θ=angle between weight and feature.

Hands-on Demo: Face Verification (1:1)

Goal

Implement a minimal face recognition system using a pretrained embedding model and cosine similarity.

Task

1. Upload three face images:
   - two images of the **same person**
   - one image of a **different person**
2. Compute embeddings using a pretrained model (FaceNet).
3. Compute cosine similarity between embeddings.
4. Choose a threshold and implement a **verification decision**:

$$\text{same person} \quad \text{if} \quad s > \tau$$

Notebook
Hands-on Face Embeddings (Colab notebook)

Concept illustrated
- embedding representation
- cosine similarity
- threshold-based verification (1:1)

Evaluation Protocols and Metrics

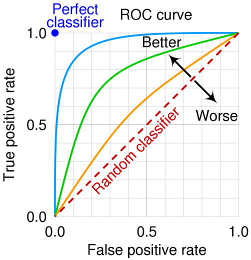

Figure: ROC-style evaluation for verification.

Two main tasks: Verification (1:1) and Identification (1:N)

Verification metrics

- **FAR** — false acceptance rate
- **FRR** — false rejection rate
- **TAR@FAR** — true accept rate at a fixed low FAR
- **EER** — equal error rate

Identification metrics

- **Rank-1 accuracy**
- **CMC curve**

Benchmarks

- **LFW:** important historical benchmark
- **IJB-B / IJB-C:** more realistic unconstrained evaluation
- **NIST FRVT:** large-scale operational testing

Section 5. Conclusion & Discussion

Conclusion & Discussion

Main technical lesson

- Face recognition is primarily an **embedding learning** problem solved with similarity-based decisions.
- Modern systems usually rely on **normalized embeddings + angular-margin training + thresholded matching**.

What matters in practice

- evaluation protocol: verification vs identification,
- operating point: security vs usability,
- image quality: blur, pose, occlusion, low resolution,
- deployment constraints: fairness, privacy, spoofing, regulation.

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